Exploration of FELLOW trait dataset

The objective of this document is to:

- visually explore the trait database
- understand trait coverage / data gaps
- check for possible inconsistencies

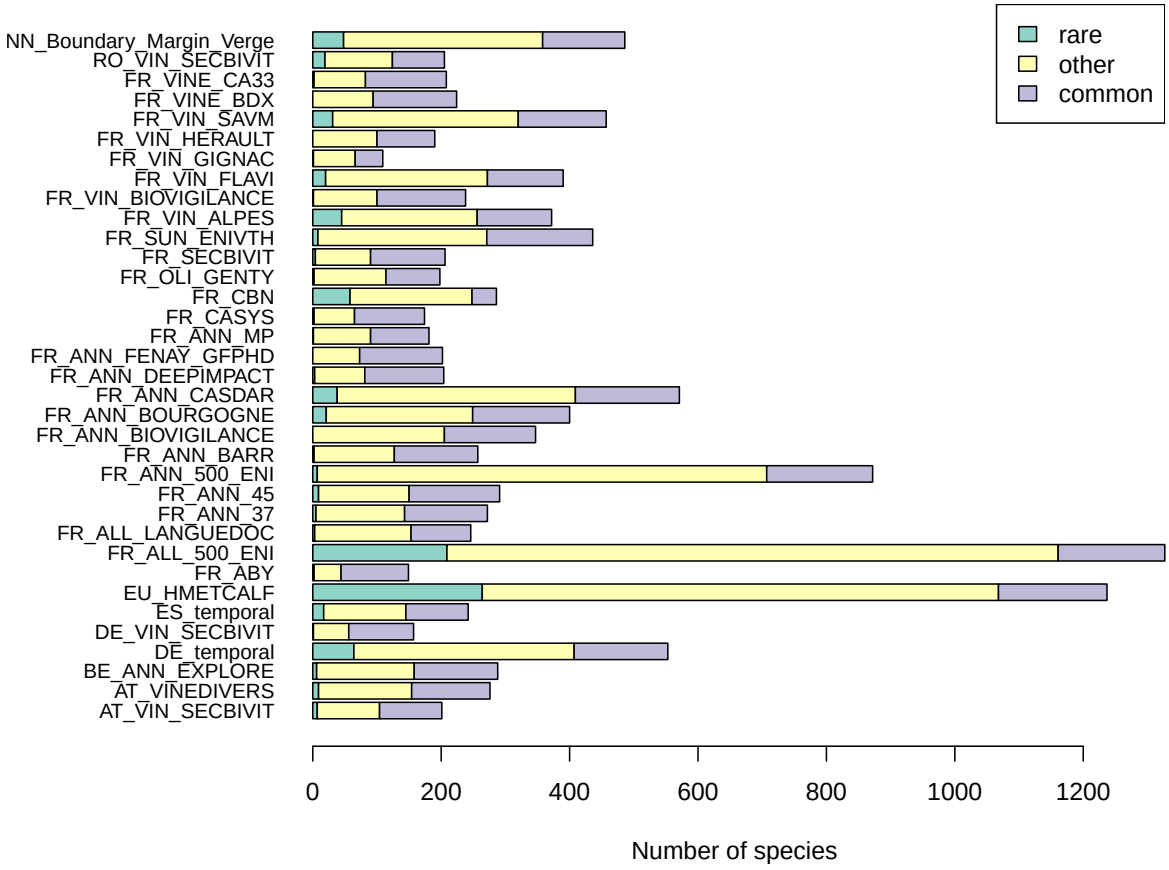
Description of the species list

We compiled the species lists from 35 datasets. After cleaning and harmonization, there were 2416 unique taxa.

```
FAMILY      GENUS    SPECIES SUBSPECIES      TRIBE    VARIETY
    18        328       1875        170          1         24
```

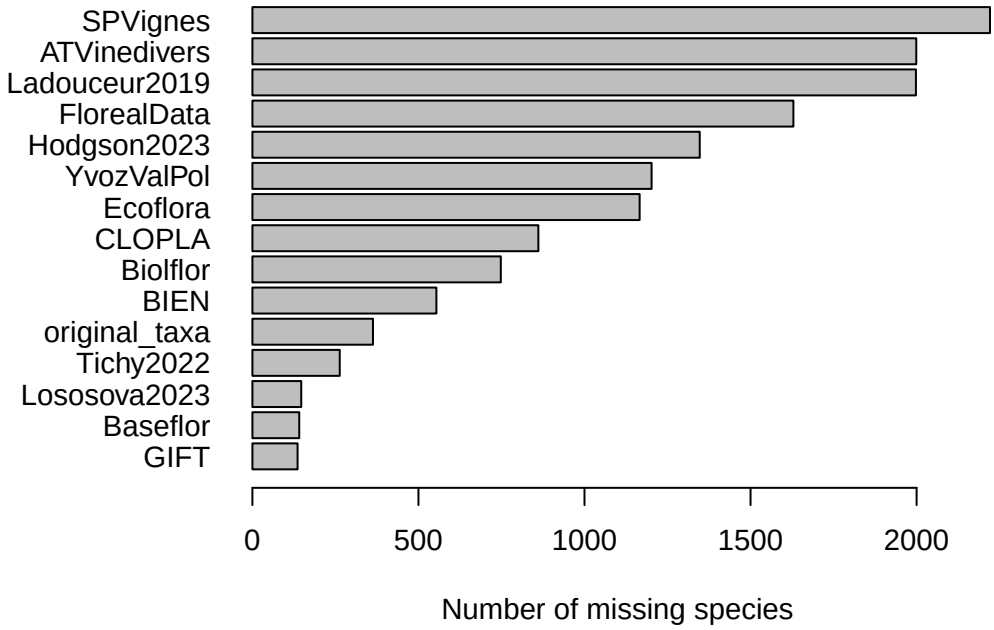
Let's define:
rare: a taxa that occurred only in a singe database
common: a taxa that is listed in 50% of the databases (18 out of 35)

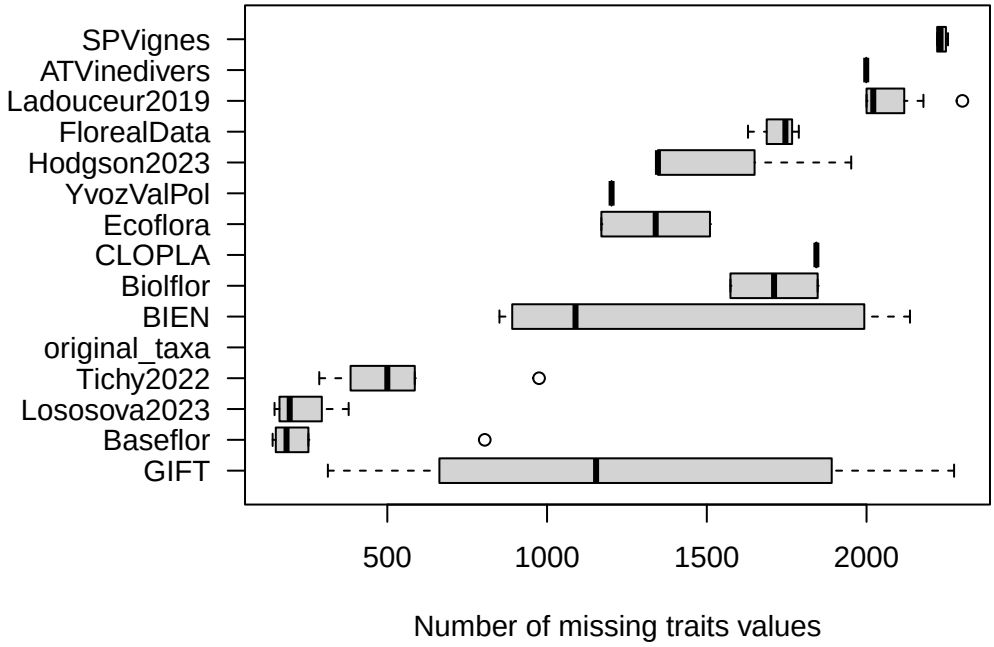
```
sp_class
  rare  other common
   909   1336    170
```

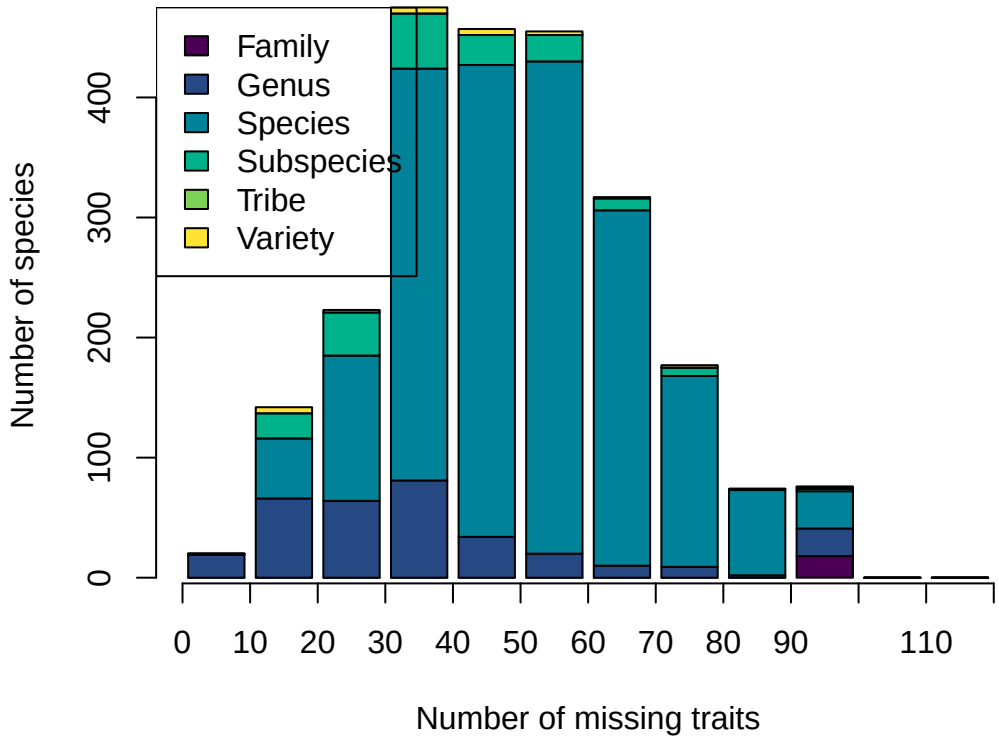


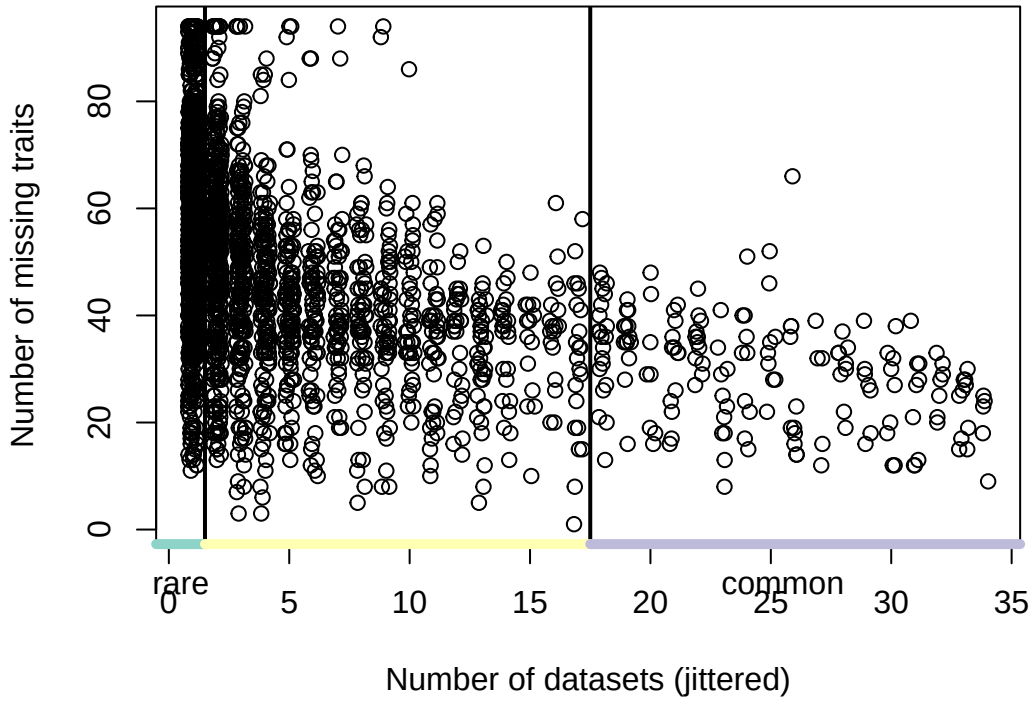
Description of trait databases

So far, we compiled 94 traits for 2416 taxa gathered from 15 trait databases. But there are many missing values.






Family
Genus
Species
Subspecies
Tribe
Variety
Number of species
0
100
200
300
400
0
10
20
30
40
50
60
70
80
90
110
Number of missing traits


Number of missing traits
0
20
40
60
80
rare
common
0
5
10
15
20
25
30
35
Number of datasets (jittered)

Taxa with no or limited trait information (N=53).

```
 [1] "Abies"
 [2] "Acacia"
 [3] "Acalypha ostryifolia"
 [4] "Agropyron"
 [5] "Amaranthaceae"
 [6] "Anagyris"
 [7] "Apiaceae"
 [8] "Arthonia arthonioides"
 [9] "Asparagaceae"
[10] "Aster"
[11] "Boraginaceae"
[12] "Brachythecium rutabulum"
[13] "Brassicaceae"
[14] "Bromeliaceae"
[15] "Bryum dichotomum"
[16] "Cardueae"
[17] "Caryophyllaceae"
[18] "Chaenomeles x superba"
[19] "Chrysanthemum"
[20] "Circaea"
[21] "Cochlearia"
[22] "Cupressus"
[23] "Elymus campestris x repens"
[24] "Ephippiger perforatus"
[25] "Geraniaceae"
[26] "Imbribryum subapiculatum"
[27] "Lamiaceae"
[28] "Lavandula"
[29] "Liliaceae"
[30] "Lilium"
[31] "Marchantia polymorpha subsp. ruderalis"
[32] "Matthiola"
[33] "Milium"
[34] "Moehringia"
[35] "Muhlenbergia"
[36] "Orchidaceae"
[37] "Orchis"
[38] "Paronychia"
[39] "Pelargonium"
[40] "Piptatherum"
```

```
[41] "Poaceae"
[42] "Primulaceae"
[43] "Psoralea"
[44] "Rhizogemma staphylina"
[45] "Riccia"
[46] "Riccia sorocarpa"
[47] "Riccia warnstorfii"
[48] "Roemeria apula"
[49] "Rosaceae"
[50] "Rubiaceae"
[51] "Thapsia"
[52] "Ulmaceae"
[53] "x Triticosecale"
```

Open question:

How to deal with families taxa?

How to deal with missing trait values?

Summary of trait completness

Trait	N database	N taxa	Completness (%)
Dispersal.mode	4	2331	96
Plant.height	8	2284	95
Chorology	1	2275	94
Pollination	1	2265	94
Dispersal.distance	1	2238	93
Flower.color	5	2237	93
Inflorescence	2	2237	93
Lifecycle	2	2184	90
Seed.mass	6	2135	88
Ellenberg.Salinity	1	2129	88
Flowering	2	2059	85
Disturbance.Frequency	1	2053	85
Disturbance.Frequency.herblayer	1	2053	85
Disturbance.Severity	1	2053	85
Disturbance.Severity.herblayer	1	2053	85
Grazing.Pressure	1	2053	85
Mowing.Frequency	1	2053	85
Soil.Disturbance	1	2053	85
Photosynthetic.pathway	1	2039	84
Ellenberg.Light	1	2031	84
Ellenberg.Reaction	1	1998	83
Ellenberg.Nutrients	1	1834	76
Ellenberg.Moisture	1	1830	76
SLA	4	1715	71
Pollination.syndrome	3	1698	70
Flower.UV.reflectance	3	1460	60
Ellenberg.Temperature	1	1441	60
Leaf.area	2	1430	59
Leaf.dry.mass.content	1	1256	52
Floral.symmetry	1	1214	50
Flower.class	1	1214	50
Flower.type	1	1214	50
Nectar.quantity	1	1214	50
Pollen.quantity	1	1214	50
PV.Bees	1	1214	50
PV.Bumblebees	1	1214	50
PV.butterflies	1	1214	50
PV.Hoverflies	1	1214	50
Lifeform	2	845	35

Trait	N database	N taxa	Completness (%)
Flower.length	2	589	24
Clonal.index	2	581	24
lateral.spread	1	575	24
offspring	1	573	24
offspring.wsmall	1	573	24
Flowering.onset	2	553	23
Vegetative.propagation	1	464	19
Leaf.nitrogen.content	1	423	18
Growth.form	1	417	17
Plant.lifespan	1	417	17
Strategy.Grime	1	417	17
Flowering.duration	1	416	17
Seeding.onset	1	416	17
Seeding.season	1	416	17
Clonal.presence	1	384	16
Leaf.carbon.to.nitrogen.content	1	280	12
Seed.dormancy	1	238	10
Strategy	1	174	7
Flower.width	2	165	7
Radial.growth	1	116	5

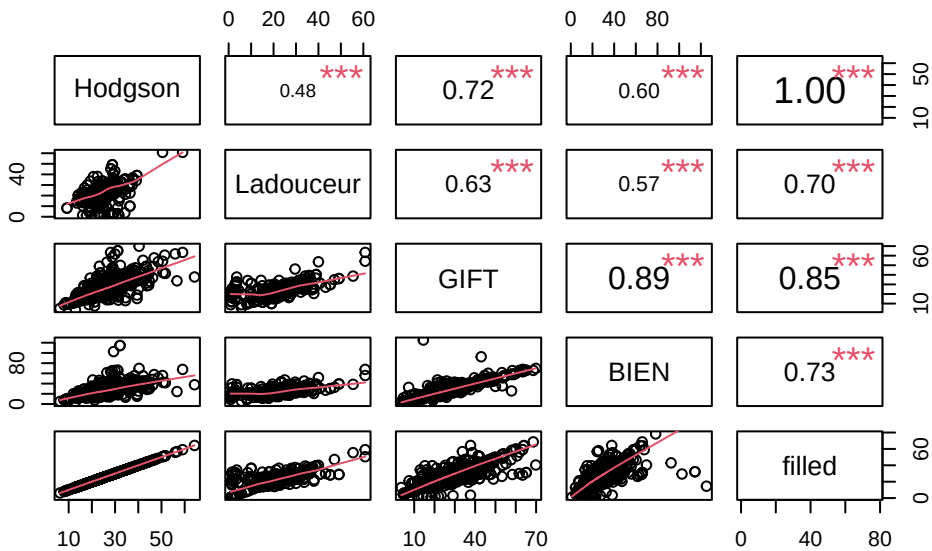
Comparison and merge

SLA

There are four sources of information for Specific leaf area (SLA) : Hodgson et al. 2023 (in mm2/mg), Ladouceur et al. 2019 (in cm2/g), GIFT (in cm2/g), and BIEN (in m2/kg = mm2/mg).

To merge and fill the SLA values (in m2/kg), we used by order of preference:

- Hodgson et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN



Number of NAs:

```
Hodgson Ladouceur      GIFT      BIEN    filled
   1347      2060      1019      1018       701
```

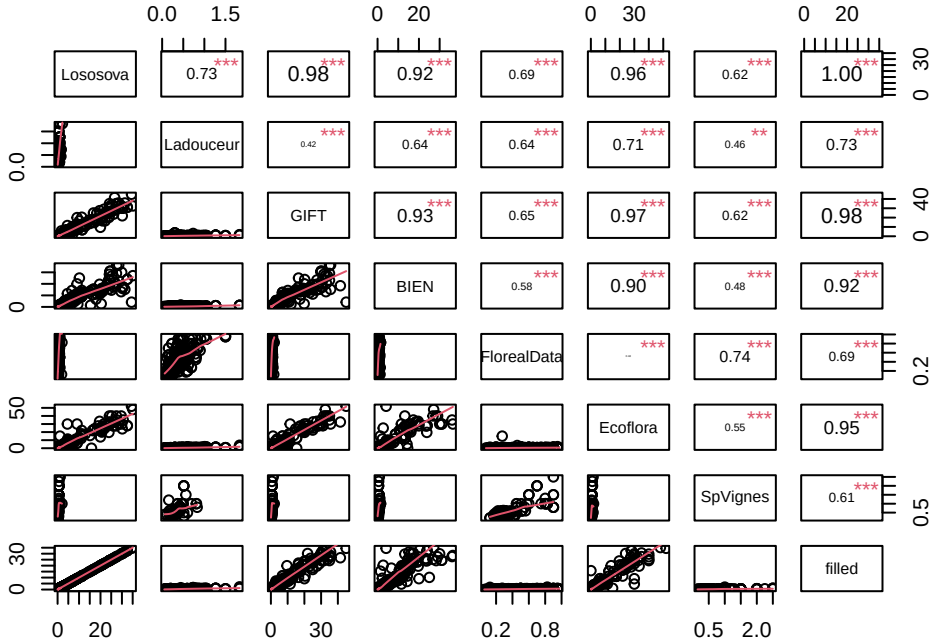
Values are highly correlated, but values from *Ladouceur et al. 2023* are only weakly correlated to other data sources (r<0.65) maybe due to different measuring protocol (?).

Plant height

There are seven sources of information for plant height. SPVignes, FlorealData, and Ladouceur_2019 are limited to small plants (<1m) (no trees) *and FlorealData>100cm must be clean.*

To merge and fill the plant height values (in m), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- FlorealData
- Ecoflora
- SpVignes



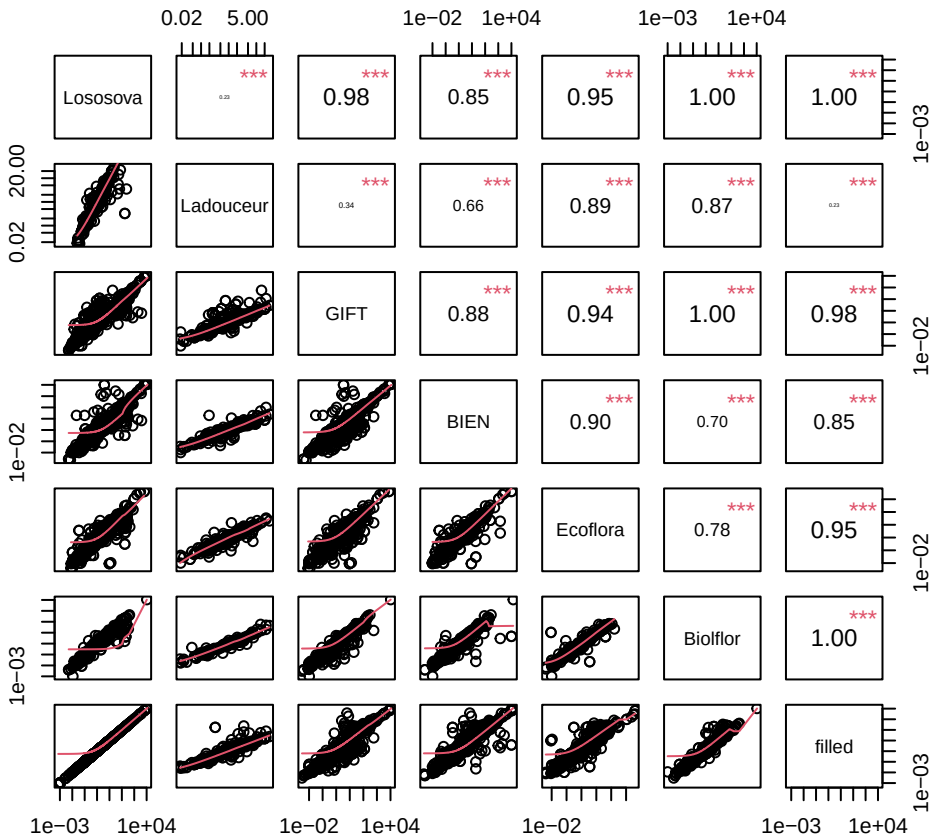
Number of NAs:

```
Lososova   Ladouceur       GIFT       BIEN FlorealData    Ecoflora
     211        2021        663        851        1885        1170
SpVignes      filled
    2255         160
```

Seed mass

There are six sources of information for seed mass. To merge and fill the seed mass plant height values (in mg), we used by order of preference:

- Lososova et al. 2023
- Ladouceur et al. 2019
- GIFT
- BIEN
- Ecoflora
- Biolflor



Number of NAs:

```
Lososova Ladouceur      GIFT      BIEN  Ecoflora  Biolflor    filled
     379      2019       433       856      1510      1847       281
```

Flower colour

There are five sources of information for flower colour, but it must be cleaned

```
   Baseflor        BIEN        GIFT FlorealData  YvozValPol
        253        2096        2093        1746        1202


baseflor
             Blanc       Blanc, jaune Blanc, jaune, bleu Blanc, jaune, rose
               428                 37                  5                  4
       Blanc, rose  Blanc, vert, rose               Bleu        Bleu, blanc
                56                  1                172                 11
 Bleu, blanc, rose        Bleu, jaune  Bleu, jaune, rose         Bleu, rose
                12                  5                  2                 14
             Jaune        Jaune, rose        Jaune, vert             Marron
               548                  3                  1                 38
              Noir               Rose               Vert         Vert, bleu
                 2                354                306                 18
 Vert, jaune, rose         Vert, rose
                 1                 39
```